



SCP, FileZilla, WinSCP and Their Role in DevOps & Cloud Infrastructure Engineering

AWS Hands-On Lab: Secure File Transfer Practice

Lab Server IP Address: 16.147.227.8

Protocol: SFTP (SSH)

Port: 22

Authentication: Individual Username and Password provided by instructor

LAB 1: Using WinSCP (Windows Users)

- Step 1: Open WinSCP application.
- Step 2: Click 'New Site'.
- Step 3: Set File Protocol to SFTP.
- Step 4: Host Name: 16.147.227.8.
- Step 5: Port Number: 22.
- Step 6: Enter your assigned Username.
- Step 7: Enter your assigned Password.
- Step 8: Click 'Login'.
- Step 9: If prompted with SSH fingerprint warning, click 'Accept'.
- Step 10: Left panel shows your local computer. Right panel shows your remote AWS home directory.
- Step 11: To upload: Drag a file from left panel (local) to right panel (remote).
- Step 12: To download: Drag a file from right panel (remote) to left panel (local).
- Step 13: Confirm transfer success in the transfer log window.
- Step 14: Create a folder named 'lab-practice' in your home directory.
- Step 15: Upload at least 3 files into 'lab-practice'.
- Step 16: Download one file back to your Desktop.
- Step 17: Log out after completion.

LAB 2: Using FileZilla (Windows / macOS / Linux)

- Step 1: Open FileZilla Client.
- Step 2: Click File > Site Manager.
- Step 3: Click 'New Site'.
- Step 4: Protocol: SFTP - SSH File Transfer Protocol.
- Step 5: Host: 16.147.227.8.
- Step 6: Port: 22.
- Step 7: Logon Type: Normal.
- Step 8: Enter your assigned Username.
- Step 9: Enter your assigned Password.
- Step 10: Click 'Connect'.
- Step 11: Accept the SSH key fingerprint if prompted.
- Step 12: Left panel shows local files. Right panel shows your remote AWS home directory.
- Step 13: Upload a file by dragging from local panel to remote panel.
- Step 14: Download a file by dragging from remote panel to local panel.
- Step 15: Create a folder named 'downloads-test' on your local machine and download at least 2 files into it.
- Step 16: Verify file integrity by opening downloaded file.
- Step 17: Disconnect after completion.

Validation Checklist

- Successful authentication to 16.147.227.8.
- Files uploaded to your personal home directory.
- Files successfully downloaded to your laptop.
- Correct directory structure created.
- No access to other students' directories (permission denied if attempted).

Troubleshooting Guide

- If 'Connection refused': Ensure port 22 is open and SSH is running.
- If 'Authentication failed': Verify username and password.
- If 'Permission denied': Ensure you are operating inside your own home directory.
- If transfer fails: Check internet connection and retry transfer.